National Institute of Technology, Rourkela
End Semester Examination, Autumn 2022
Compiler Design (CS3007)
5^{th} semester B.Tech in Computer Science & Engineering

Duration: 3 Hours **Full Marks: 50**
Number of pages: 1

- **Attempt all questions.**
- **All parts of a question should be answered together.**

Q 1. Three languages are required for designing a new compiler - source language (X), target language (Y) and implementation language (Z). Symbolically it is represented as C_Z^{XY}. Consider A and B are assembly languages for machines M_A and M_B, respectively. Using a language S design a compiler C_A^{LA} for a new language L that will produce target code for M_A. Then using M_A, L and C_A^{LA} design a cross-compiler C_A^{LB} for L which will run on M_A but produce target code for M_B. Write the sequence of design using C_Z^{XY} notation. [6]

Q 2. In C language **10.25f** and **10.25** are instances floating point and double constants, respectively. They can also be represented in scientific notation (**1.025e+1**). Design a lexical analyzer to recognize floating point and double constants for C language. [6]

Q 3. Consider the grammar G with rule $S \rightarrow (S)|id$

(a) Show that $G \in LR(0)$. [4]

(b) Give a trace for acceptance of the input string "((id)" with the help of $LR(0)$ parser for G. Report "accept" or "error". [2]

Q 4. (a) Construct a syntax directed translation (SDT) scheme to convert a hexadecimal number to decimal. [4]

(b) Use the SDT to convert $(AFB1)_{16}$ to $(44977)_{10}$. [4]

Q 5. Given an 8-bit processor with general purpose register set $R = \{r_0, r_1, r_2, r_4, r_5, r_6, r_7\}$, a set of memory variables $M = \{m_i | i \in \{1, 2, \cdots\}\}$ and a set of all possible immediate values I. The instruction set of the target machine contains the following instructions:
MOV X,Y ; $X \leftarrow Y$ *ADD X,Y ;* $X \leftarrow X + Y$ *SUB X,Y ;* $X \leftarrow X - Y$ *MUL X,Y ;* $X \leftarrow X * Y$
DIV X,Y ; $X \leftarrow X / Y$ *CMP X,Y ;* compares the values in X and Y
JGT Ll – jump to level *Ll* if $X > Y$ after the execution of a *CMP X,Y*; where $l \in \{1, 2, \cdots\}$
JMP Ll – unconditional jump to level *Ll*, where $l \in \{1, 2, \cdots\}$
For *MOV* if $X \in R$ then $Y \in R \cup M \cup I$; if $X \in M$, then $Y \in R$.
For *ADD*, *SUB*, *MUL*, *DIV* and *CMP*; $X \in R$ and $Y \in R \cup M \cup I$.
An m_i is required whenever there is a shortage of registers during code generation.

(a) Write the 3-address intermediate code the following code fragment. [8]

```
if(a>10)
    c=a–b/2;
else
if(a>5)
    c=4*a+b;
else
    c=2;
```

(b) Generate target code from the intermediate code using labeled binary tree method. Use the given instruction set. The number of registers allocated should be minimum. Avoid usage of memory variables. [8]

Q 6. For the following code fragment:

```
for (int i = 0; i < n; i++) {
  x = y + z;
  a[i] = 6 * i + x * x;
}
```

(a) Find the loop-invariant expressions and induction variables. [4]

(b) Perform code motion and strength reduction. [4]

*******All the best*******